

Dynamic Knowledge Graph Construction

Benchmarking Report:

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1. Introduction & Objective

The primary objective of this project is to design an software pipeline capable of transforming unstructured text related to our own ontology into structured RDF triples.

This report benchmarks a baseline architecture, utilizing Named Entity Recognition combined with rule-based relation extraction, to highlight the limitations of deterministic boundaries and justify the transition to advanced ML models.

2. The Baseline Pipeline

Ontology: My ontology is based on the Star Wars semantics, defining core classes (Character, Place, Weapon) and their explicit relations (locatedIn, hasWeapon, relativeOf) with strict domain and range constraints.

Dataset: A synthetic dataset comprising 10 tiered-complexity Star Wars stories, generated via LLM and noted in JSONL format.

Architecture: Entity detection is handled by GLiNER utilizing dynamic zero-shot label matching. Relation candidate extraction relies on exhaustive constraints (e.g., linking all Characters to all Places found within the same sentence -> luke skywalker and leia organa).

3. Results & Evaluation

The baseline pipeline was evaluated against the synthetic JSONL dataset. The performance metrics are as follows:

Metric	Value
True Positives (Correct)	13
False Positives (Wrong)	22
False Negatives (Missed)	16
Precision	37.14%
Recall	44.83%
F1-Score	40.62%

4. Error Analysis

High False Positives (The Heuristic Problem): The primary cause of the low precision (37.14%) is the massive over-linking generated by rule-based extraction. For instance, if a text states "Luke and Leia went to Tatooine," the heuristic rule blindly generates the triple [Leia, hasWeapon, Luke's Lightsaber]. While logically valid according to the ontology's domain/range constraints, it is contextually false, because Leia has no such weapon.

Symmetric Property Errors: The heuristic rule for the symmetric property `relativeOf` assumes that any characters mentioned in the same sentence are related, artificially inflating False Positives.

Recall Issues (False Negatives): The pipeline missed 16 triples (44.83% Recall) primarily because complex sentence structures bypass simple iterative loops, or GLiNER misses implicit entity mentions.

5. Proposed Improvements (Advanced Architectures)

To mitigate the severe over-linking issues inherent to heuristic extraction, the pipeline must transition to Architecture 3 or 4, as outlined in the course materials. The heuristic relation builder should be replaced by a **Supervised Transformer model** (such as a fine-tuned BERT or RoBERTa for Sequence Classification).

In this advanced paradigm, the model evaluates the contextual validity of candidate entity pairs against the ontology. Instead of blindly associating all entities within a sentence, the Transformer classifies whether a specific relation (e.g., `hasWeapon`) is contextually True or False. This approach significantly reduces False Positives while preserving the dynamic entity recognition capabilities of GLiNER.